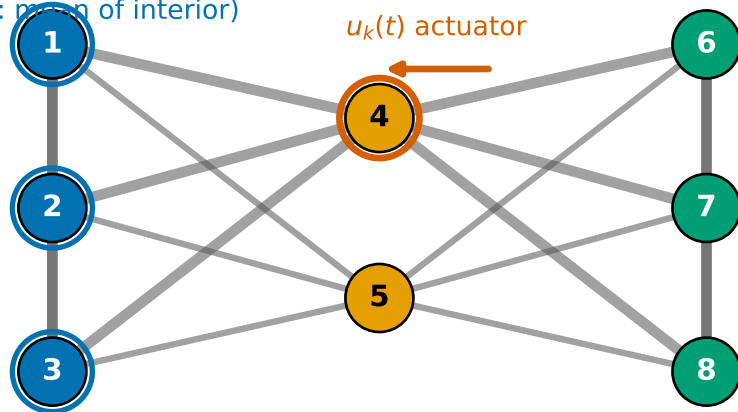


## Steering Figure 1 — What are we controlling?

What we are steering

$$Y = \frac{1}{3}(x_1 + x_2 + x_3)$$

(target: mean of interior)



Controlled dynamics:

$$dX_t = -A_q X_t dt + e_k u(t) dt + \sqrt{2} dW_t$$

$$A_q = (I + Q_q) \Omega_0$$

Key fact (verified, Part 2B):

$\Sigma(t) \equiv \Sigma_0$  for ANY open-loop  $u(t)$

so the instantaneous Gaussian blanket

$$I(X_{1:3}; X_{6:8} | X_{4,5}) = 0$$

stays exactly zero while the MEAN is being steered toward a target  $Y^*$ .

Only the mean trajectory is driven;  
individual stochastic realizations are  
NOT claimed to hit the target exactly.